Water of Crystallisation in Hydrated Copper(II) Sulfate

Grade 11 Chemistry | S1.4 Counting particles by mass | Practical worksheet

Name: _____ Date: _____ Lab partner: _____

Aim and background

Aim: To find x in the formula $CuSO_4 \cdot xH_2O$ by heating the hydrated salt to constant mass.

Blue hydrated copper(II) sulfate contains water molecules locked into its crystal lattice: its **water of crystallisation**. Gentle heating drives this water off and leaves white anhydrous copper(II) sulfate. The mass lost is the mass of water. Converting the masses of water and anhydrous salt to moles gives the ratio $n(H_2O) : n(CuSO_4)$, which is x.

$$\mathbf{CuSO_4 \cdot xH_2O(s) \rightarrow CuSO_4(s) + xH_2O(g)}$$

Safety and equipment

Safety: Wear goggles throughout. Copper(II) sulfate is harmful if swallowed and toxic to aquatic life, so put all residues in the labelled waste container, not down the sink. Do not overheat: above about 650 °C the anhydrous salt decomposes, giving black CuO and toxic sulfur oxides. Handle the hot crucible only with tongs.

Equipment	Uncertainty
Electronic balance (2 d.p.)	± 0.01 g
Crucible with lid, tongs	–
Pipe-clay triangle, tripod, heatproof mat, Bunsen burner	–
Hydrated copper(II) sulfate, about 3 g, and a spatula	–
Glass stirring rod (optional, to break up lumps)	–

Method

1. Weigh the clean, dry crucible and lid together.
2. Add about 3 g of hydrated copper(II) sulfate. Spread it evenly over the base so it heats evenly, then weigh the crucible, lid and sample.
3. Put the crucible on the pipe-clay triangle with the lid slightly ajar, so water vapour can escape. Heat gently with a medium blue flame.
4. Keep heating gently until all the solid has turned from blue to white. If it starts to turn grey or black, lower the flame at once: this means CuO is forming.
5. Turn off the Bunsen burner. Close the lid and let the crucible cool on the heatproof mat for about 5 minutes.
6. Weigh the crucible, lid and anhydrous solid.
7. Heat gently for 2 more minutes, cool with the lid on, and weigh again. Repeat until two masses in a row agree to within 0.01 g (constant mass).
8. Optional: add a few drops of water to the cooled white solid and note what you see and feel.

Raw data

Record every mass to 2 decimal places, matching the balance's precision.

Measurement	Mass / g (± 0.01 g)
A: crucible + lid (empty)	
B: crucible + lid + hydrated salt	
C1: crucible + lid + anhydrous salt, 1st weighing	
C2: 2nd weighing	
C3: 3rd weighing (if needed)	
C: final constant mass	

Qualitative observations (colour before and after, condensation, what happened in step 8):

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Processing the data

Use $M(CuSO_4) = 159.62\ g\ mol^{-1}$ and $M(H_2O) = 18.02\ g\ mol^{-1}$ from the data booklet. Treat their uncertainties as negligible.

Uncertainty rules: When you add or subtract, add the **absolute** uncertainties. When you multiply or divide, add the **percentage** uncertainties.

Step	Quantity	Working	Value	Absolute unc.	% unc.
1	mass of hydrated salt / g	B − A		± 0.02 g	
2	mass of anhydrous $CuSO_4$ / g	C − A		± 0.02 g	
3	mass of water lost / g	B − C		± 0.02 g	
4	$n(CuSO_4)$ / mol	step 2 ÷ 159.62			same as step 2
5	$n(H_2O)$ / mol	step 3 ÷ 18.02			same as step 3
6	$x = n(H_2O) \div n(CuSO_4)$	step 5 ÷ step 4			step 4 + step 5

To convert % to absolute uncertainty: absolute unc. = (% unc. ÷ 100) × value

x = _____ ± _____ Formula: $CuSO_4 \cdot$_____H_2O

Percentage of water by mass in your sample = (step 3 ÷ step 1) × 100 = _____ %

Percentage error compared with the accepted value x = 5 = _____ %

Conclusion and evaluation

1. Does x = 5 fall inside your uncertainty range? What does this tell you about whether your error is mainly random or systematic?

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2. Calculate the theoretical percentage of water by mass in $CuSO_4 \cdot 5H_2O$, and compare it with your experimental value.

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3. Which mass has the largest percentage uncertainty? Suggest two ways to reduce it.

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4. Why must you heat to constant mass, and why is the crucible cooled with the lid on? (Hint: anhydrous $CuSO_4$ is used to test for water.)

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5. Complete the table: does each error make the calculated value of x **too high** or **too low**? Explain why.

Error	x (high / low)	Reason
(a) Some blue solid is left after heating		
(b) The solid is overheated and turns black		
(c) Solid spits out of the crucible during heating		
(d) The anhydrous salt absorbs moisture while cooling		

6. **Extension:** Magnesium sulfate forms $MgSO_4 \cdot 7H_2O$. A student heats 2.46 g of it to constant mass and is left with 1.20 g. Calculate x, and suggest why $MgSO_4$ gives no colour change to tell you when the water has gone.

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